BIOL3100: Introduction to Data Analysis for Biologists

Fall 2026

Instructor Information

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Student Hours: Tuesday 11:15 AM -1 PM or by appointment

* Please do not use Teams messages for communication. Messages sent through Canvas may experience delays

Course Description

This course provides an introduction to analyzing data in the R software environment, assuming no previous R experience.
We cover:

- the justification of using code to explore and analyze data
- best practices for dealing with data
- experimental design, modeling, and hypothesis testing
- how to create publication-quality figures that show interesting relationships in our data sets
- project organization and proper reporting

You need to bring your own laptop to class every day. That way you have installation privileges and you will be set up for the future as you continue to analyze data.

Learning outcomes

Upon successful completion, students should be able to…

1. Demonstrate proficiency in proper data entry, management, and storage for scientific research with an emphasis on reproducibility.
2. Convert untidy data to "tidy data" for analyses.
3. Discuss the basic principles of exploratory data analyses within appropriate software environments.
4. Evaluate the rationale behind using code to analyze data and present results.
5. Develop computational skills for processing common biological data formats, such as DNA sequences.
6. Create appropriate and meaningful data visualizations using appropriate software environments.
7. Integrate principles of experimental design, statistical modeling, hypothesis testing, and data visualization to critically analyze a unique data set.

8. Present a fully-reproducible report using a unique data set.

Class Meetings, Times and Location

Tuesday & Thursday 10:00 AM – 11:15 PM
Science Building (SB) 149

Course Materials
We will be using datasets provided by the instructors or contributed by students. Additional information and resources will be shared as the course progresses, depending on the direction we take. No textbook is required for this course. However, the following supplemental reading sources may be helpful:

- Big Book of R: https://www.bigbookofr.com
- R for Data Science: https://r4ds.had.co.nz
- Introduction to Data Science: http://rafalab.dfci.harvard.edu/dsbook/

How to be successful in this class?
This course requires significant outside work… meaning that if you aren't practicing at least 6 hours a week *outside of class*, then you are going to have a really rough time of it. Like, you'll fall behind and will be frustrated and scared and sad. Don't let this happen. Practice! Come to my office! Ask questions! And spend time **every day** using R. Seriously. Every day.

The course website has external resources and practice tasks for every week. Use these to make the most of this opportunity. **Data fluency is currently one of the most desirable job skills**, including in biology. Most undergrad science students do not learn these skills, putting you at a major competitive advantage if you make the effort to learn this.

Computers are not a passing fad. Don't get left behind. – Dr. Geoffrey Zahn

Student evaluation and assignments
Grading Scheme
We will use a point system. The points you have accumulated by the end of the course determine your grade as follows:

Points	Letter Grade
700-800	A
640-699	B
560-639	C
480-599	D
<480	E

Points are based on:

- 10 Assignments - 20 pts each
- 4 Skills Tests - 100 pts each
- 1 Final Project - 200 pts

Assignments
These are shown on the [course website](https://gzahn.github.io/data-course/) (https://gzahn.github.io/data-course/). They (along with all necessary data and files) are also available in their respective directories on the [GitHub repository](https://github.com/gzahn/Data_Course) (https://github.com/gzahn/Data_Course). They will generally consist of requiring you to complete a task using R and to upload your code as an R script to Canvas and/or GitHub. Some may vary. These assignments will not be accepted for credit after the due date.

Skills Tests
These will be similar to the sorts of tasks on the precedent assignments, and they will be open-source (you can use notes, internet, etc.). They focus on completing some data analysis tasks.

Final Project
Beginning early in the semester, we will decide on individual projects based on personal interest. Working with instructor feedback, you will come up with a question that interests you and will identify a data set that can address that question. You will then apply the data exploration and visualization skills you learn in class to prepare a well-formatted report that contains all the code and results of your analyses. You are encouraged to use your own data if you have any. Most importantly this project will require you to teach yourself new R skills using the resources we learn about (or any others). You will be required to learn analyses explicitly not covered in this course (Check the course website to see what we are covering), and to apply them correctly to your analysis and report.

Examples of unique topics could include, but are not limited to:

- Time-series analyses
- Genomic Profiling
- New statistical testing methods
- Metagenomic assembly
- Learning a new R package
- Unique visualization methods

Your topic/question is due before Skills Test 1
Soon after, you will work with the instructor to identify a suitable publicly-available data set (or use your own)
Then you will work with the instructor to identify a new skill to be applied for your project
Your project will culminate in a reproducible report that integrates your background research, code, results, figures, and discussion into a single reproducible document on a personal website that you will create

Individual versus Group Work
All work is to be done individually unless it is clearly identified as group work. By default all assignments are not group work. This means that you cannot work on assignments with others, even if your spouse, friends or family, and turn them in as your own. A "group" in this context means more persons than yourself. You also cannot have others complete the work for you or in part on your behalf.

Course Schedule

This schedule is subject to change. Please make sure to look for the latest syllabus on canvas.

Week	Date	Topic
Week 1		
R	Aug 20	Course Introduction, Installing Software, Command-line
Week 2		
T	Aug 25	Git and GitHub
R	Aug 27	File paths
Week 3		
T	Sep 1	Getting to Know R- R Data types and conversions
R	Sep 3	**No Class- Dr. Liang out for conference**
Week 4		
T	Sep 8	Getting to Know R- Packages and Projects
R	Sep 10	**No Class- UVU close**
Week 5		
T	Sep 15	Visualizing a Data Set
R	Sep 17	Visualizing a Data Set
Week 6		
T	Sep 22	Clean and Transform Data
R	Sep 24	Clean and Transform Data
Week 7		
T	Sep 29	Data Wrangling
R	Oct 1	Data Wrangling
Week 8		
T	Oct 6	Writing Functions
R	Oct 8	Conditional Execution
Week 9		
T	Oct 13	Model Building and Testing
R	Oct 15	**No Class- Fall Break**
Week 10		
T	Oct 20	Model Building and Testing
R	Oct 22	Model Building and Testing
Week 11		
T	Oct 27	R-Markdown
R	Oct 29	Reproducible Reports
Week 12		
T	Nov 3	**No Class- Dr. Liang out for conference**
R	Nov 5	Proper Project Organization
Week 13		
T	Nov 10	Proper Project Organization
R	Nov 12	Building a website
Week 14		

T	Nov 17	Building a website
R	Nov 19	Data Analysis from raw to report
Week 15		
T, R	Nov 23-27	**No Class- Thanksgiving Break**
Week 16		
T	Dec 1	Data Analysis from raw to report
R	Dec 3	Final Presentation

Course Late Work:
All assignments must be turned in at the times indicated on Canvas or in the syllabus. Late work will not be accepted without a doctors or lawyers note unless previously discussed with and approved by your instructor. Computer problems, work schedule, lack of internet access, traffic, travel, or similar do not constitute rationale for late homework. Your instructor has the right to modify their handling of late work.

AI Syllabus Statement

In this course, you are encouraged to explore various tools and resources, including generative AI, to enhance your learning experience. AI can be a valuable supplement for understanding complex concepts, brainstorming, and debugging code. However, learning is a personal journey, and true skill development requires active, individual effort. When utilizing AI tools, use them as an aid to your own work rather than a replacement for it. Avoid simply copying and pasting AI-generated content, as doing so hinders your long-term growth and understanding.

To maintain academic integrity, you must not submit scripts or code directly generated by AI. All submitted work must be actively reviewed, revised, and cross-referenced with course materials to ensure it truly reflects your own knowledge. If a submitted assignment or script is obviously AI-generated, a first offense will result in a formal warning, while any subsequent offense will result in a grade of zero. Additionally, you may be required to explain your work or redo assignments and exams in person to prove individual mastery of the material.

Academic Honesty:
At Utah Valley University, faculty and students operate in an atmosphere of mutual trust. Maintaining an atmosphere of academic integrity allows for free exchange of ideas and enables all members of the community to achieve their highest potential. Our goal is to foster an intellectual atmosphere that produces scholars of integrity and imaginative thought. In all academic work, the ideas and contributions of others must be appropriately acknowledged and UVU students are expected to produce their own original academic work.

Faculty and students share the responsibility of ensuring the honesty and fairness of the intellectual environment at UVU. Students have a responsibility to promote academic integrity at the university by not participating in or facilitating others' participation in any act of academic dishonesty. As members of the academic community, students must become familiar with their rights and responsibilities. In each course, they are responsible for knowing the requirements and restrictions regarding research and writing, assessments, collaborative work, the use of study aids, the appropriateness of assistance, and other issues. Likewise, instructors are responsible to clearly state expectations and model best practices.

Further information on what constitutes academic dishonesty is detailed in UVU Policy 541: Student Code of Conduct.

Accommodation:
The Americans with Disabilities Act (ADA) and Section 504 of the Rehabilitation Act of 1973, as amended, prohibit Utah Valley University from engaging in discrimination on the basis of disability in any program or activity. Discrimination is also prohibited in all aspects of employment against persons with disabilities who, with reasonable accommodation, can perform the essential functions of a job. Students who believe they have been denied program access or otherwise discriminated against because of a disability are encouraged to initiate a grievance by contacting the Accessibility Services Director, Sherry Page at 801-863-8747. Employees can contact the ADA Coordinator, Irene Whittier at 801-863-8389.

Upon request, this information is available in alternative formats, such as mp3, Braille, or large print. To request this format, email asd@uvu.edu.

Students needing accommodations due to a disability including temporary and pregnancy accommodations may contact the UVU Accessibility Services at accessibilityservices@uvu.edu or 801-863-8747. Accessibility Services is located on the Orem Campus in LC 312.

Title IX Statement:
Title IX makes it clear that violence and harassment based on sex and gender (which includes sexual orientation and gender identity/expression) is a civil rights offense subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories such as race, national origin, color, religion, age, status as a person with a disability, veteran's status or genetic information. If you or someone you know has experienced or experiences harassment or sexual assault including, dating and domestic violence, stalking or sexual exploitation, you are encouraged to report it to the Title IX Coordinator in the Office for Equal Opportunity and Affirmative Action, BA-203, (801) 863-7999.

Please be aware that all faculty members and university employees are considered "Responsible Employees" and are required to report incidents of sexual misconduct and relationship violence and thus cannot guarantee confidentiality. Please know that you can seek confidential resources at UVU Student Health Services, SC-221, (801) 863-8876. Please visit https://www.uvu.edu/equalopportunity/ for more information.

Religious Accommodations:
UVU values and acknowledges a wide range of faiths and religions as part of our student body, and as such provides accommodations for students. Religious belief includes the student's faith or conscience as well as the student's participation in an organized activity conducted under the auspices of the student's religious tradition or religious organization. The accommodations include reasonable student absences from scheduled examinations or academic requirements if they create an undue hardship for sincerely held religious beliefs. For this to occur, the student must provide a written notice to the instructor of the course for which the student seeks said accommodation prior to the event.
The UVU campus has a place for meditation, prayer, reflection, or other forms of individual religious expression as is described on their website.